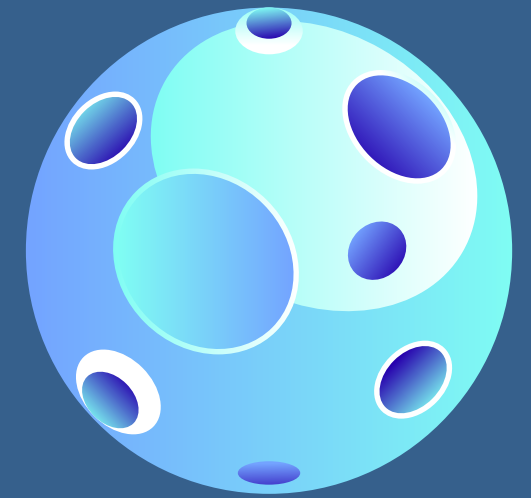


MACROECONOMIC DETERMINANTS OF INDONESIA'S GDP GROWTH (2010–2023)

Nadine Kholifah Umaroh

BACKGROUND



Analysis of Macroeconomic Determinants of Indonesia's Economic Growth Using a Supervised Learning Linear Regression Approach (2010–2023)

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ABSTRACT

This study analyzes the macroeconomic determinants of Indonesia's economic growth using a supervised learning multiple linear regression approach for the period 2010–2023. The model examines the effects of inflation, real interest rate, unemployment rate, and current account balance on GDP growth using annual secondary data and Ordinary Least Squares (OLS).

The results show that all variables have theoretically consistent relationships with GDP growth. The real interest rate is identified as the most dominant determinant (Beta = -1.3036), followed by current account balance (-0.9223), inflation ($+0.7209$), and unemployment rate (-0.1674). The model explains 49.97% of GDP growth variation ($R^2 = 0.4997$), indicating moderate explanatory power.

These findings highlight the critical role of monetary policy in influencing economic growth, although the model has limitations in capturing extreme shocks such as the COVID-19 pandemic.

Keywords: Economic Growth, Linear Regression, Macroeconomic Variables, Indonesia, Supervised Learning

PROBLEM:

1. Indonesia's growth drivers are debated
2. Especially post-COVID, when standard relationships may have shifted

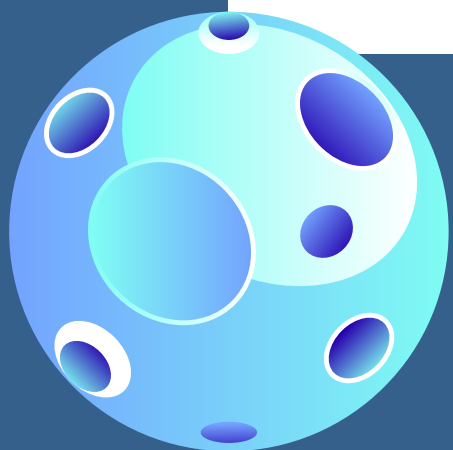
THE QUESTIONS:

- Among inflation, interest rates, unemployment, and trade balance — which one actually drives GDP growth the most?
- Does monetary policy outweigh real-sector factors?

TASK:

- Quantify how inflation, real interest rate, unemployment, and current account balance drove GDP growth (2010–2023)
- Supervised learning regression approach

TECH: Python · Scikit-learn · Pandas · OLS Regression



PROCESS



- Cleaned 14 years of annual macroeconomic data (Pandas)
- Built & validated OLS regression (Scikit-learn LinearRegression)
- Standardized coefficients (Beta) to compare variable impact directly



RESULT

Variance Explained (R^2): 49.97%

Real Interest Rate (β): -1.30 — Dominant Factor

Nearly half of Indonesia's year-to-year GDP growth swings (2010–2023) are explained by just these 4 variables and real interest rate has the single strongest pull on growth among them.

This points to monetary policy not fiscal spending or labor market conditions as Indonesia's most powerful lever for managing growth. The model's blind spot: it can't account for shocks like COVID-19, where growth swings came from forces outside any of these four variables

